

JOINS AND SUBQUERIES

table 1 : Accounts

A_id	A_holder

table 2 : transaction

Amt	acc_id	transac_id	Date

Q/A given these two tables, give
id, name, num of transaction
(March 2024)

total
value
per day

For joins:

- * Need to have similar columns
- * Some times header may not be same but interpretation should be same

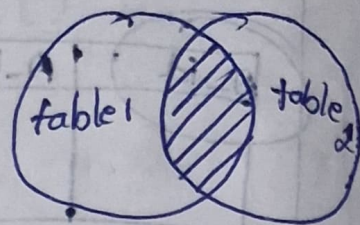
→ For these Q/A

- ① Need to apply join ✓
- ② What kind of join to use?

Inner join

→ Major focus to find the intersection of two tables

→ i.e. Common values



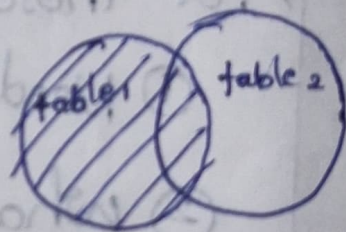
By default: JOIN --> refers to INNER JOIN.

For eg.: give the ^{only} records when two column (acc-id, A-id) matches.

Left join

→ to find common fields also (when two columns gets matched)

→ But looking for all the values that available in table 1



For eg: (Inner Join, Left join)

Accounts table

A_id	A-holder
2138	Monisha
2148	Priya
2157	Dharshini
3148	Selvi
5174	Gayathri

transaction table

date	acc-id	tran-id	Amt
12-02-24	2138	4171	2000
13-02-24	2138	4175	4000
13-02-24	5174	4127	5000
13-03-24	2157	4108	7000

Q/A With the given table, give the table with Account id, Account-holder-name, Number of transaction (March 2024), total-value WITH Inner join (It only display the records when two column matches)

Account id	Account-holder-name	Num of transaction	total value
2138	Monisha	2	6000
2157	Dharshini	1	7000
5174	Gayathri	1	5000

With left join

→ It also give the records of the column matches

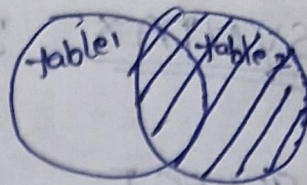
→ It also gives all values in table 1
(id)

Account id	Account holder name	Num of transaction	total value
2138	Monisha	2	6000
2140	Priya	null	null
2157	Dharshini	1	7000
3148	Selvi	null	null
5174	Crayatheni	1	5000

→ It will print table 1 entirely. Doesn't matter whether there is a common id available in table 2 or not.

RIGHT JOIN

Opposite to Left join



CROSS JOIN (OR) FULL OUTER JOIN

Results of every join

✓ CROSS JOIN → Cartesian prod of two tables

% ⇒ Multiplication of num of rows in both table

Table A

5 rows

Table B

10 rows

It gives all the possible combinations

O/p : 50 rows

FULL OUTER JOIN

Return all rows from both table, with null values (NULL) where no matches exists

SYNTAX JOINS

SELECT ^{columns} FROM table 1
^{by default inner} (JOIN) table 2 ^{right table}

ON table 1. column = table 2. column

↓
similar col in both table

INTERVIEW QUESTION (Actual)

Google

SELECT A-id, A-holder, COUNT(Transactions)
SUM(Amt) FROM Accounts

JOIN Transaction
ON Account.A-id = Transaction.Acc-id
WHERE month (Transaction-date) = 03
group by Acc-id, A-holder;

SELF JOIN

(Leetcode 181)

Employee Earning More than their Managers

→ Apply Alias name in Self JOIN

SELECT emp1.name as Employee

FROM Employee emp1

JOIN Employee emp2

ON emp1.managerId = emp2.id

AND emp1.salary > emp2.salary

Employee

emp.managerId

id	name	Salary	managerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	Null
4	Max	90000	Null

emp2.id
then

Query which course has highest enrollment rate
FIRST JOINS (completely)

select Learners, selected Course, CourseName
COUNT(*) as enrollment-count from
Courses JOIN Learners
ON CourseId = Learners. SelectedCourse
GROUP BY selectedCourse
ORDER BY enroll-ment-count
LIMIT BY 1;

Computationally
Expensive

join for less of rec
USING JOINS with SUB QUERY

→ First limit the record then
join with Course table.